

Experiment 1 – Mathematical Pendulum

Group 7
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We hereby confirm that this lab report is entirely our own. Where we have consulted the work of others this is always clearly stated. All statements taken literally from other writings or referred to by analogy are marked and the source is always given. This paper has not yet been submitted to another examination office, either in the same or similar form.

Abstract

The aim of this experiment was to measure the local gravitational acceleration in Bremen using a simple pendulum and to see how the period changes with different deflection angles. We tried two methods first measuring the timing of the swings manually with a stopwatch, and then measuring automatically with a light barrier connected to the Cassy system. The pendulum length was measured as $L = (2.5145 \pm 0.001) \text{ m}$. From the stopwatch method, the period was $T = (3.179 \pm 0.003) \text{ s}$, which gave gravitational acceleration $g = (9.940 \pm 0.062) \text{ m/s}^2$. With the light barrier, we got $g = (10.026 \pm 0.063) \text{ m/s}^2$. Both values matched the standard local value of 9.813310 m/s^2 within the error range, though the light barrier was more accurate and consistent than the stopwatch. The experiment also investigated the relationship between the period and the angle of deflection. For small angles 10° or below, the period stayed the same, which confirms the small-angle approximation. At larger angles, the period increased, and our results fit the expected trend. In short, the experiment showed that pendulum is a reliable way to measure gravitational acceleration, and it also confirms the relationship between period and angle of deflection.

Keywords: Mathematical Pendulum, Local Gravitational Acceleration, Experimental Lab Report

Introduction

A mathematical pendulum is a mass m suspended from a fixed point by a massless, inextensible string of length L . When displaced by a small angle θ and released, the pendulum undergoes periodic motion under the influence of gravity. When the pendulum swings back and forth, the mass moves along an arc of a circle with radius L , which is the length of the string. Polar coordinates can also be used to describe this motion. As in straight-line motion, velocity is used, for the pendulum, angular velocity ω can describe the speed the pendulum is moving through its arc:

$$\omega = \frac{d\theta}{dt} = \frac{v}{L}$$

Eq. 1.1 – Angular Velocity

In this situation, θ shows the position of the pendulum's mass, and v is its tangential speed along the arc. To find the angular acceleration, we take the derivative of the angular velocity. Similarly, linear acceleration comes from differentiating the tangential velocity. These connections can be written as:

$$a = \frac{d^2\theta}{dt^2}$$

Eq 1.2 – Angular Acceleration

$$a = \frac{d^2s}{dt^2}$$

Eq 1.3 – Linear Acceleration

The gravitational force on the pendulum can be broken into two components: A radial force along the string (which only changes the tension, not the motion). A tangential force that pulls the pendulum back toward its starting position. This tangential force acts like a restoring force, similar to a spring, always pointing back toward the equilibrium position. Because of this, the pendulum swings back and forth in simple harmonic motion. The formula for the restoring force can be expressed as:

$$F(\theta) = -mg * \sin\theta$$

Eq. 1.4 – Restoring Force

Using Newton's second law $F = ma$ for the tangential force, we get:

$$F_{tang} = m * a_{tang} = m * \frac{dv_{tang}}{dt} = m * L \frac{d\omega}{dt} = m * L \frac{d^2\theta}{dt^2}$$

Eq. 1.5 – Tangential Force

When we set these to expressions for the restoring force equal each other, it can be solved to get:

$$-mg \sin \theta = m * L \frac{d^2\theta}{dt^2} \Rightarrow \frac{d^2\theta}{dt^2} + \frac{g}{L} \sin \theta = 0$$

Eq. 1.6 – Motion of a Pendulum

For small oscillations of the pendulum, the restoring torque is proportional to $\sin \theta$. Since the motion is limited to small angles ($0^\circ < \theta \leq 10^\circ$) the approximation $\sin \theta \approx \theta$ can be applied. With this simplification, the equation of motion reduces to:

$$\frac{d^2\theta}{dt^2} + \frac{g}{L} \theta = 0$$

Eq. 1.7 – Motion of a Pendulum for Small Angles

Which is the standard form of a simple harmonic oscillator. The angular frequency is obtained as:

$$\omega = \sqrt{\frac{g}{L}}$$

Eq. 1.8 – Angular frequency

$$T = 2\pi \sqrt{\frac{L}{g}}$$

Eq. 1.9 – Period of Oscillation

These equations demonstrate that the period of oscillation depends only on the pendulum's length L and the gravitational acceleration. For small angular deflections, the period is independent of the initial angle, which confirms that the pendulum behaves as a simple harmonic oscillator. Indicating that the restoring torque is directly proportional to the displacement angle. The motion of the pendulum can also be described using the law of conservation of energy. At any point during the swing, the total mechanical energy is given by:

$$\frac{1}{2}mv^2 + mgh = E(0) = mgh_{max}$$

Eq. 1.10 – Total Mechanical Energy

Where h_{max} is the maximum vertical displacement corresponding to the highest point of the swing, and h is the instantaneous height of the pendulum mass. By rearranging and simplifying this energy relation, one can derive an expression that links the period T of the pendulum to the maximum deflection angle. This shows how, for larger amplitudes, the period slightly increases compared to the ideal small-angle approximation.

$$T = 2\pi\sqrt{\frac{L}{g}} \left[1 + \left(\frac{1}{2}\right)^2 \sin^2\left(\frac{\theta_{max}}{2}\right) + \left(\frac{1 \cdot 3}{2 \cdot 4}\right)^2 \sin^4\left(\frac{\theta_{max}}{2}\right) + \dots \right]$$

Eq. 1.11 – Full Small Angle Approximation

For small angles of deflection, the formula can be approximated into:

$$T = 2\pi\sqrt{\frac{L}{g}} \left[1 + \frac{1}{16}\theta_{max}^2 \right] = T_0 \left[1 + \frac{1}{16}\theta_{max}^2 \right]$$

Eq. 1.12 – Small Angle Approximation

In this equation, T represents the actual period, while T_0 is the ideal period. To determine how the period relates to the maximum angle of deflection we can take the natural logarithm of both sides:

$$\frac{T}{T_0} - 1 = \frac{1}{16}\theta_{max}^2$$

Eq. 1.14 – Derivation of Small Angle Approximation

Taking the logarithm gives us:

$$\ln\left(\frac{T}{T_0} - 1\right) = \ln\left(\frac{1}{16}\theta_{max}^2\right) = -4\ln(2) + 2\ln(\theta_{max})$$

Eq. 1.15 – Natural Log of Small Angle Approximation

The illustration for calculating θ_{max} is outlined below:

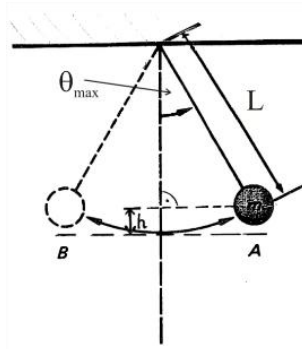


Fig. 1 – Illustration of Mathematical Pendulum

The variable Δh represents the difference in height between the highest point and the lowest point of the pendulum, measured from the ground:

$$\Delta h = h_{max} - h_{min} = L(1 - \cos \theta_{max})$$

Eq. 1.16 – Change in Height Based on Angle

To isolate the maximum angle of deflection, we can rearrange the formula:

$$\theta_{max} = \arccos\left(1 - \frac{h}{L}\right)$$

Eq. 1.17 – Angle in terms of Height

Experiment

Objective

The objective of this experiment is to determine the gravitational acceleration g by measuring the oscillation period of a mathematical pendulum with known length. Additionally, the experiment investigates how the period depends on the maximum deflection angle and compares two measurement techniques: manual timing using a stopwatch and automatic timing using a light barrier connected to a Cassy system.

Equipment

- Metal sphere
- Thin metal wire (string)
- Mounting device attached to ceiling
- Stopwatch
- Light barrier and Cassy sensor system
- Ruler and calipers (for measuring length and diameter)
- Protractor (for angle estimation)
- Measuring stand or scale

All measurements were conducted with careful attention to minimizing angular amplitude to ensure the validity of the small-angle approximation ($\sin \theta \approx \theta$). The length L of the pendulum was measured from the suspension point to the center of the sphere, and all devices were aligned to minimize parallax errors.

All measurements were performed using a simple pendulum setup consisting of a metal sphere suspended from a thin wire attached to a fixed support. The pendulum was allowed to oscillate freely in a single vertical plane. The setup made it possible to measure the oscillation period precisely and to study how it depends on the pendulum's length and deflection angle.

Calculations with **Eq. 1.17** were used to get the deflection angles, while a ruler and caliper were used to measure the wire length and the sphere's diameter with millimeter accuracy. The effective length of the pendulum was taken from the suspension point to the center of mass of the sphere. Before each trial, the pendulum was carefully aligned to avoid unwanted sideways motion or twisting of the wire.

For each measurement, the pendulum was displaced by less than 10° and released without any initial push to ensure motion under the small-angle approximation ($\sin \theta \approx \theta$). The time for a number of complete oscillations was measured using Manual measurement with a stopwatch, and Automatic measurement with a light barrier connected to a Cassy system. In both cases, several repeated measurements were performed to reduce random errors and obtain reliable mean values. The gravitational acceleration g was then calculated from the measured period T and the pendulum length L using **Eq. 1.9**. Uncertainties were determined using two approaches, statistical analysis (mean value and standard deviation) for repeated measurements, and error propagation for the calculated quantities, based on the uncertainties in the length (ΔL) and period (ΔT).

Task 1 – Measuring Gravitational Acceleration: Preparation

Measuring the diameter

The diameter of the metal sphere was measured with a caliper, and its uncertainty was noted. This value was needed to calculate the pendulum's total length.

Determining the length of the pendulum

The distance was measured by measuring the length of the string and getting the radius of the sphere by measuring its diameter and dividing by 2.

Deriving the formula for g

This was done using **Eq. 1.9** where Period, Length, and Gravitational Acceleration was present.

Best method for measuring the period

Both the stopwatch and light barrier methods were considered. The light barrier was preferred as it reduces human reaction errors and gives more consistent readings

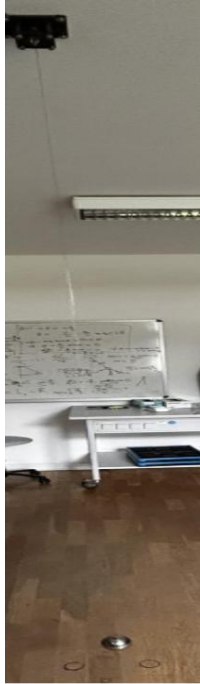


Fig. 2 – Image of Pendulum Set-Up

Task 2 – Measuring gravitational acceleration using a stopwatch

Measuring t_{10}

To measure the Period T , the experiment required the measurement of t_{10} 10 oscillations on a stop watch. These 10 oscillations were then divided by 10 to get one period T . This measurement was done 20 times to reduce random errors. When releasing the pendulum, no additional force was given and was released at an angle below 10° for harmonic motion to be observed.

Best value and Reliability

Treating the data by getting the mean and error yielded a reliable value.

Period T and its error

$$\bar{T} = \frac{\overline{t_{10}}}{10}$$

Eq. 2.1 – Period T from t_{10}

$$\Delta \bar{T} = \frac{\Delta \overline{t_{10}}}{10}$$

Eq 2.2 – Error of the Mean T from Error of the mean t_{10}

Compute g from data

Using Eq. 1.9

Error propagation (Δg from ΔT and ΔL)

$$\frac{\partial g}{\partial T} = -\frac{8\pi^2 L}{T^3} = -\frac{2g}{T}, \quad \frac{\partial g}{\partial L} = \frac{4\pi^2}{T^2} = \frac{g}{L}$$

Eq. 2.3 – Partial Derivatives of g

$$\Delta g = \sqrt{\left(\frac{\partial g}{\partial T} \Delta T\right)^2 + \left(\frac{\partial g}{\partial L} \Delta L\right)^2}$$

Eq. 2.4 – Propagated Error of g

Task 3 – Measuring Gravitational Acceleration Using a Light Barrier

Measuring the period

The pendulum was set to swing with a small deflection angle (less than 10°). Each time, it was released carefully without any push, and the light barrier connected to the Cassy system automatically recorded the time for one complete oscillation. The measurement was repeated 20 times.

Calculating g and its uncertainty

The uncertainty was determined with the same formulas, **Eq. 1.9**, **Eq. 2.3**, and **Eq. 2.4**

Task 4 – Comparison of Results

Compare the gravitational acceleration with each other and with the standard gravitational acceleration in Bremen North, their uncertainty ranges, and discuss.

5. Functional Relationship between Period and Angle of Deflection

Determine $h_{\min}$

This was done by measuring the sphere's height from the ground to its bottom.

Calculate height for 40° angle of deflection

This was done through **Eq. 1.17**

Task 5 – Measure period T at 40° deflection angle 5 times and calculate the mean and error

This was done with light barrier and data recorded in a table.

Measure period t_s at 40° deflection angle and calculate one T

This was done with light barrier and the individual values of each period measured by the Cassy was summed up.

Measure period T at different release heights/angles of deflection

Using **Eq. 1.17** the heights were first set and their corresponding angles calculated, this was to reduce uncertainty in release height. Release height was also measured by the bottom of the sphere. This experiment was measured with Cassy, and so caution was used so as to not hit it. The data was recorded on a table.

Make a plot of $T(\theta_{\max})$, T as the y value and $\theta_{\max}$ as the x value

This was done through Excel's data visualization tools.

Make a plot of dependence, plotting $\ln\left(\frac{T}{T_0} - 1\right)$ with $\theta_{\max}$, getting the linear regression.

This was done through Excel's data visualization tools.

Results and Data Analysis

In this section, the results and data will be presented. This will answer the question of quantifying the local gravitational acceleration in Bremen.

The First Step was to measure the diameter of the sphere and length of the pendulum and their errors. This measurement yielded:

$$D = 5.5 \pm 0.1 \text{ cm} \quad \text{and} \quad L = 251.45 \pm 0.1 \text{ cm}$$

The length of the pendulum given by the TA was $254.9 \pm 0.1 \text{ cm}$, longer than what was measured. However, the measured length of the pendulum ended at the center of the sphere and not the bottom.

Once the length L of the pendulum was ascertained, the experiment to measure gravitational acceleration by Stop Watch was done.

Table 1.1 – Gravitational Acceleration Measured Stop Watch

	$t_{10} - s$
1	31.35
2	31.40
3	31.34
4	31.44
5	31.26
6	31.69
7	31.58
8	31.68
9	31.92
10	31.51
11	31.47
12	31.35
13	31.45

Table 1.2 – Mean, Standard Deviation, Error, By and Propagated Error of t_{10} and T for Stop Watch

t_{10} – Mean	31.5155 s
t_{10} – Standard Dev.	0.164876 s
t_{10} – Error	0.036867 s
T – Mean	3.15155 s
T – Standard Dev.	0.016488 s
T – Error	0.003687 s
ΔT	0.01 s
ΔL	0.1 cm
ΔG	0.063426211 m/s ²

14	31.46
15	31.42
16	31.39
17	31.71
18	31.58
19	31.71
20	31.60

Using the mean period for 10 oscillations of the pendulum, we obtained the Period for one by dividing by 10. With the Period and Length, the gravitational acceleration calculated:

$$T = 3.152 \pm 0.004 \text{ s}$$

$$g = 10.026 \pm 0.063 \text{ m/s}^2$$

The next experiment was the measurement of gravitational acceleration using a light barrier. The equipment used was a Cassy, and it measured one period instead of measuring 10.

Table 2.1 – Gravitational Acceleration Measured By Light Barrier

	T - s
1	3.177
2	3.177
3	3.177
4	3.183
5	3.183
6	3.183
7	3.179
8	3.179
9	3.178
10	3.178
11	3.178
12	3.181
13	3.179
14	3.179
15	3.179
16	3.175
17	3.175
18	3.175
19	3.181
20	3.181

Table 2.2 – Mean, Standard Deviation, Error, and Propagated Error of T for Light Barrier

T – Mean	3.17885 s
T – Standard Dev.	0.002539685 s
T -Error	0.000567891 s
ΔT	0.01 s
ΔL	0.1 cm
ΔG	0.061929441 m/s ²

With the Period and Length, the gravitational acceleration calculated:

$$T = 3.179 \pm 0.003 \text{ s}$$

$$g = 9.940 \pm 0.062 \text{ m/s}^2$$

Comparing the gravitational acceleration calculated in these experiments with the standard local gravitational acceleration in Bremen North, $9.813310 \pm 0.000041 \text{ m/s}^2$ (Donfack, Kammer, & Mohaghegh, Classical Physics Lab Manual, 2025), and accounting for error range, the measured value corroborates the standard gravitational acceleration.

After deriving the gravitational acceleration, the second part of the experiment proceeded. The relationship between the Period and the Angle of Deflection was determined. This was done through dropping the pendulum from different heights, angles of deflection, and measuring the period through the light barrier.

The angle of deflection had a range of 10° to 40° . 10° being the maximum for the small angle approximation. These angles were converted into heights, $h_{\min} = 24 \pm 0.1 \text{ cm}$ for when the pendulum was at 0° and $h_{\max} = 82.8 \pm 0.1 \text{ cm}$ for when the pendulum was at 40° .

Table 3.1 – Period T at 40° Angle of Deflection

	T – s
1	3.279
2	3.276
3	3.278
4	3.278
5	3.279

Table 3.2 – Period t_s for 40° Angle of Deflection

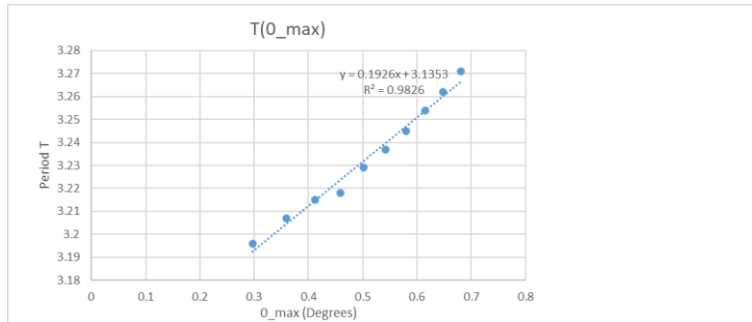
t_5 – s	16.462
t_1 – s	3.2924

Table 3.3 – Mean, Standard Deviation, and Error of T at 40° Angle of Deflection

T – Mean	3.278 s
T – Standard Dev.	0.0012247 s
T -Error	0.0005477 s

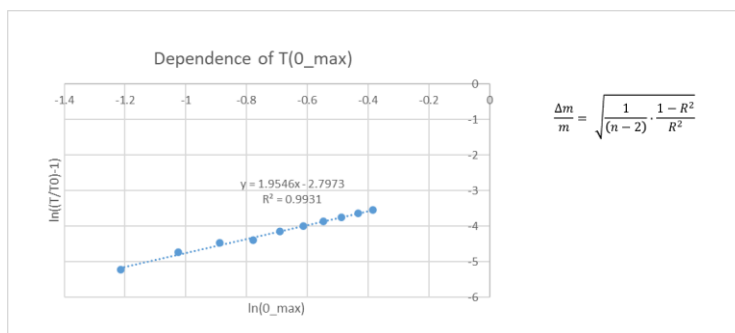
Table 3.4 – Relationship Between Period and Angle of Deflection

T	$h_{\max}$	Δh	$\theta_{\max} \text{ Rad}$	$\theta_{\max} \text{ Deg}$	$\ln \theta_{\max}$	$(T/T_0) - 1$	$\ln((T/T_0) - 1)$
3.271	80	56	0.6804468	38.986729	-0.3850057	0.0289885	-3.54085709
3.262	75	51	0.6481925	37.138694	-0.4335676	0.0261573	-3.64362857
3.254	70	46	0.6145013	35.208334	-0.4869442	0.0236406	-3.74478866
3.245	65	41	0.5791179	33.181014	-0.5462491	0.0208094	-3.87234988
3.237	60	36	0.5417059	31.03746	-0.6130321	0.0182928	-4.0012489
3.229	55	31	0.5018065	28.751394	-0.6895407	0.0157761	-4.14925626
3.218	50	26	0.4587658	26.285346	-0.7792154	0.0123158	-4.39687435
3.215	45	21	0.4115937	23.582582	-0.8877186	0.011372	-4.47659783
3.207	40	16	0.3586571	20.549536	-1.0253886	0.0088554	-4.72672742
3.196	35	11	0.2968805	17.009997	-1.2144257	0.005395	-5.2222766



Graph 1.1 – $T(\theta_{\max})$ Period as a function of Theta

This illustrates the relationship between period and the angle of deflection, showing a high correlation. As the Angle of Deflection increases, the Period increases. The correlation, however, should be exponential.



Graph 1.2 – Dependence of $T(\theta_{\max})$ Linear Correlation of $T(\theta_{\max})$ with Error equation

The expected slope for this experiment was 2 and the graph yielded a 1.9546 ± 0.0816 correlation. Indicating that the experiment was successful in determining the relationship between the Period and Angle of Deflection.

Error Analysis

Statistical Error Analysis

As seen in **Tables 1.2, 2.2, and 3.2** there were statistical errors borne from systematical and random errors.

The Mean, Standard Deviation, and Error of the mean were calculated using their formulas. Subsequently, the error propagation was also calculated for Δg using **Eq. 2.3** and **Eq. 2.4**.

There is also the error of the slope when calculating the relationship. As seen in **Graph 1.2** the error equation was used to find the error of the slope.

Human Errors

Human errors also contribute to the result. These errors include the measurement of lengths and heights, time on stop watch, and errors on the angle of deflection.

Error due to Friction

Friction on the pendulum parts and air resistance has an effect, though, it is miniscule. And so, it was not included in the calculations.

Discussion

Preparation

Preparing for the experiment was getting the Length L parameter of our experiment. The L measured was compared with the given L by the TA and was shorter than it. This value was used the most as it is used in finding Gravitational Acceleration with Period, and the heights for angle of deflection. However, this, second to the operation of a stopwatch, had a great chance for human error. This human error could be mitigated by making the string, instead of wire, a rod. A rod ensures that the length of the pendulum is more easily measured because it is not prone to stability issues. Additionally, when dropping, the string can veer, due to human error, and is at risk of hitting and damaging the light barrier.

Gravitational Acceleration using Stopwatch

The method using a stopwatch measures 10 oscillations and then divides that to get 1 period. Using the small angle approximation, the pendulum moves in a harmonic motion, so the oscillations can be attributed to gravity and its acceleration. Thus, the measured period was used to find gravitational acceleration. This method gave results further from the standard value. Errors attributed to this are human errors using stopwatch, and human errors.

Gravitational Acceleration using Light Barrier

This method seems to be more accurate as it reduces human error when operating the stopwatch. It also uses the small angle approximation and employs the mathematical pendulum's harmonic motion. The difference between this and the previous experiment is that the main error for the previous is human while the main error to this experiment is instrumental.

Comparison of Results between Standard and Experiment

The local gravity measured on the stopwatch and the light barrier was similar but not exact with the standard. This is because of errors in operating the stopwatch and the limitations of the set-up. Other variables are weather and vibrations. The values were close though, having overlapping error ranges, with both values higher the standard. Additionally, the gravitational acceleration measured by the light barrier was more precise and closer to the standard value. The two experiments prove that Mathematical Pendulum can be used to measure gravitational acceleration that corroborates with the standard local gravitational acceleration.

Conclusion

In Task 2, the gravitational acceleration calculated was $10.026 \pm 0.063 \text{ m/s}^2$, and in Task 3, it was $9.940 \pm 0.062 \text{ m/s}^2$. These values are similar to the standard local gravitational acceleration in Bremen North, which is $9.813310 \pm 0.000041 \text{ m/s}^2$. This indicates that the Mathematical Pendulum Experiment can measure gravitational acceleration. g being directly proportional to the length and inversely proportional to the period at a set angle of deflection. The relationship between the Angle of Deflection and Period was also proven, showing a linear correlation with a slope of 1.9546 ± 0.0816 . Showing that, for angles above 10° , as the angle increases, the period increases.

References

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